Name : Divya pandey

PRN : 202501050016

Practical 4

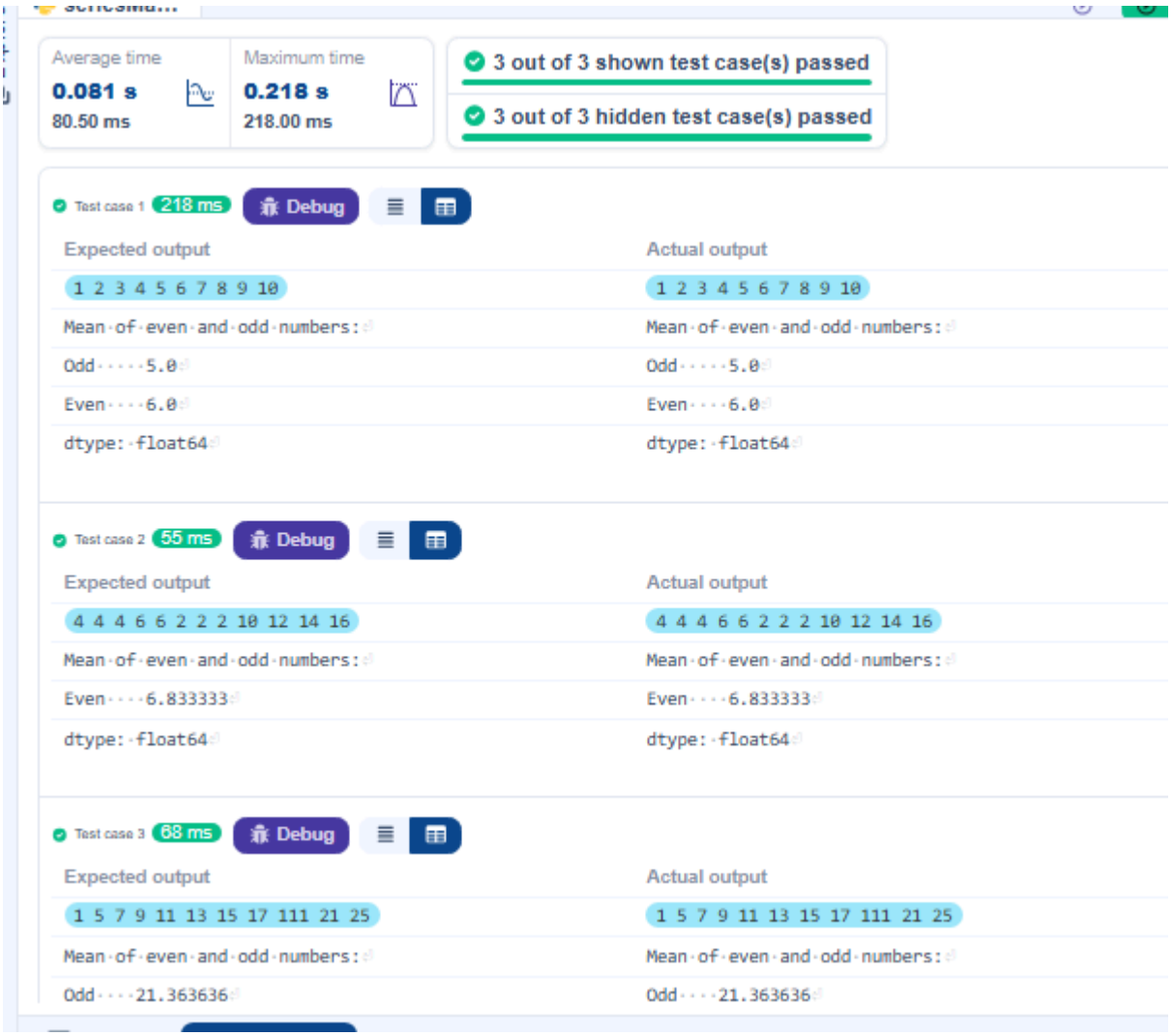
---

1)

Program :

```
import pandas as pd

# Take inputs from the user to create a list of numbers

numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers

series = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean

grouped = series.groupby(series % 2 == 0).mean()

# Display the mean of even and odd numbers with labels

grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

print("Mean of even and odd numbers:")

print(grouped)
```

Output :



2)

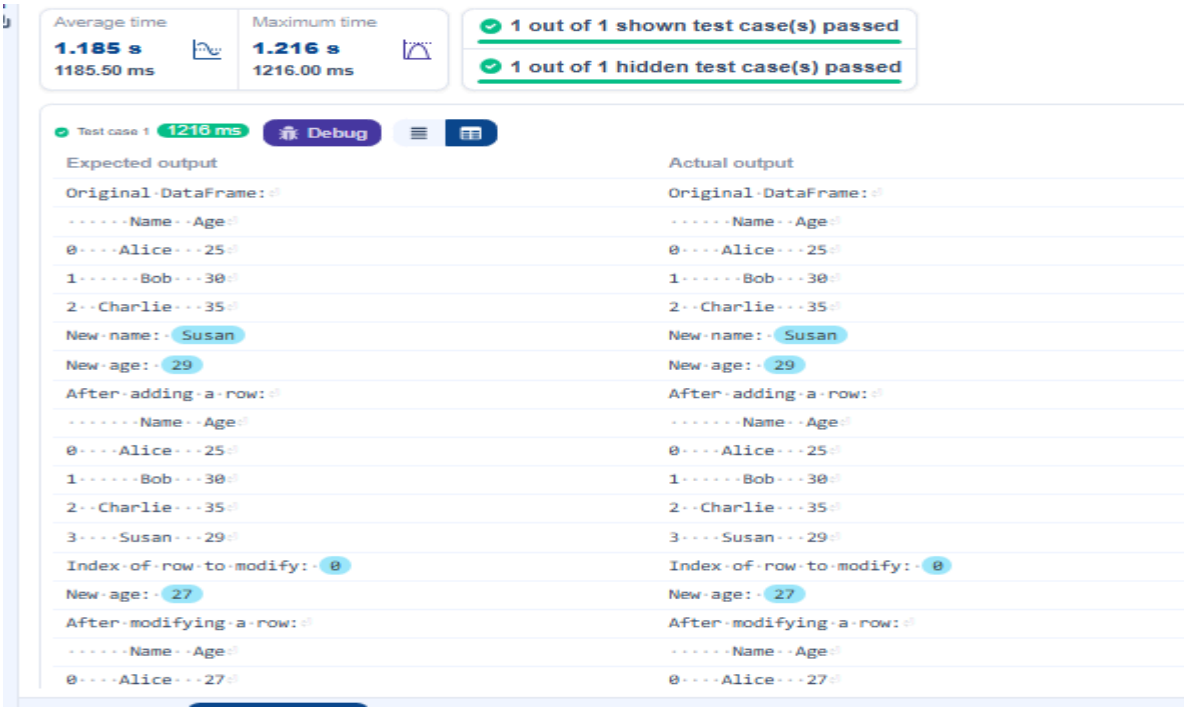
Program :

```
import pandas as pd


# Provided dictionary of lists

data = {

    'Name': ['Alice', 'Bob', 'Charlie'],

    'Age': [25, 30, 35],

}
```

```
# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)

# Adding a new row
new_name=input("New name: ")
new_age=int(input("New age: "))
new_row={'Name':new_name,'Age':new_age}
df=df.append(new_row,ignore_index=True)

# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
modify_row_index=int(input("Index of row to modify: "))
new_ag_value=int(input("New age: "))
df.at[modify_row_index,'Age']=new_ag_value

# Display the DataFrame after modifying a row
print("After modifying a row:")
print(df)

# Deleting a row
```

```
delete_row_index=int(input("Index of row to delete: "))
df=df.drop(delete_row_index).reset_index(drop=True)
# Display the DataFrame after deleting a row
print("After deleting a row:")
print(df)

# Adding a new column
genders=input("Enter genders separated by space: ").split()
df['Gender']=genders

# Display the DataFrame after adding a new column
print("After adding a new column:")
print(df)

# Modifying a column
df['Name']=df['Name'].str.upper()
# Display the DataFrame after modifying a column
print("After modifying a column:")
print(df)

# Deleting a column
df=df.drop(columns=['Age'])
# Display the DataFrame after deleting a column
print("After deleting a column:")
print(df)
```

Output :



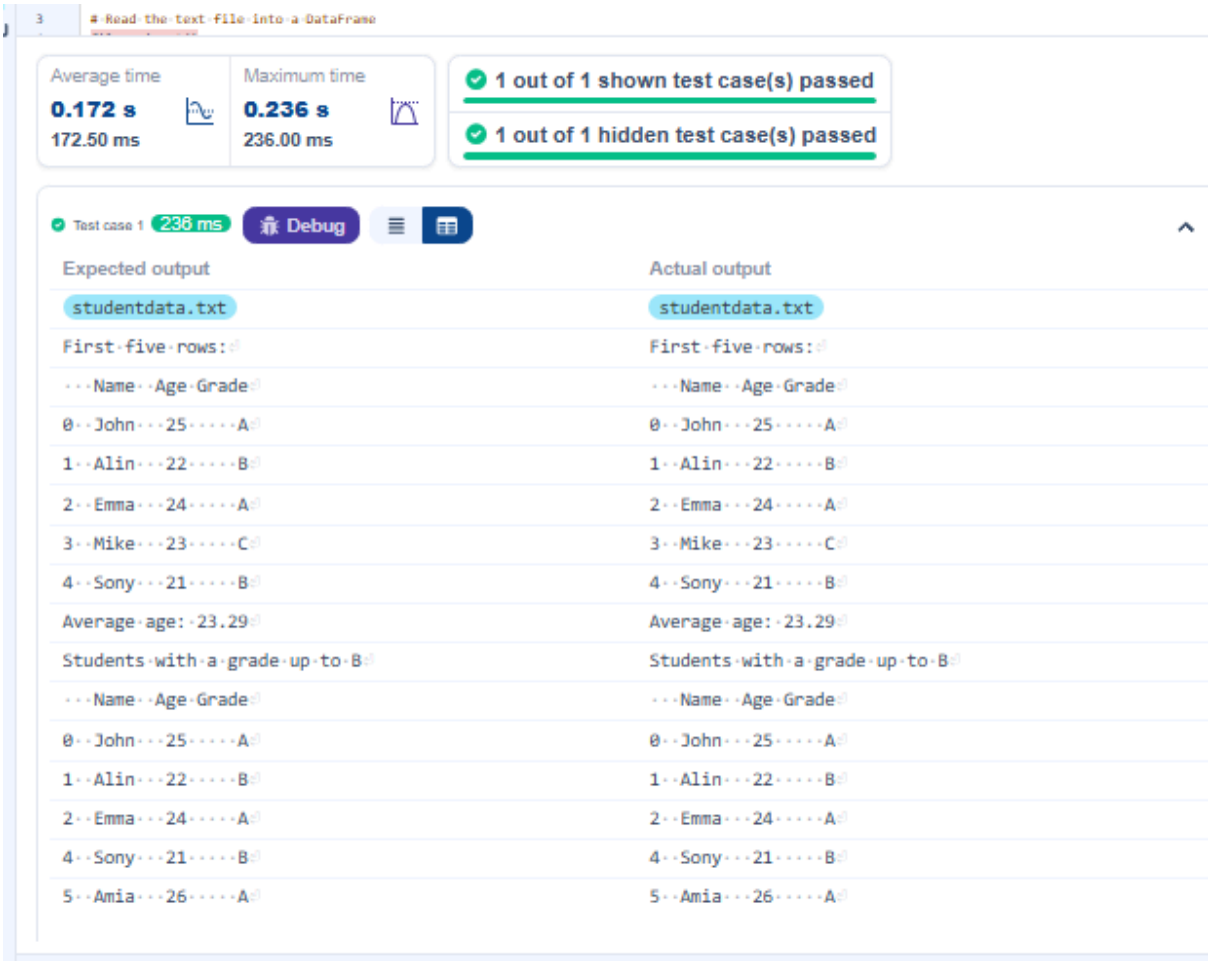
3)

Program :

```
import pandas as pd


# Read the text file into a DataFrame

file = input()

data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])


# Display the first five rows

print("First five rows:")

print(data.head())


# Calculate the average age of sstudents(limited to 2 decimal places)

average_age = round(data["Age"].mean(),2)
```

```
print(f"Average age: {average_age}")


# Filter students with a grade up to 'B'

filtered_students = data[data["Grade"] <= "B"]

print("Students with a grade up to B")

print(filtered_students)
```

Output :



4)

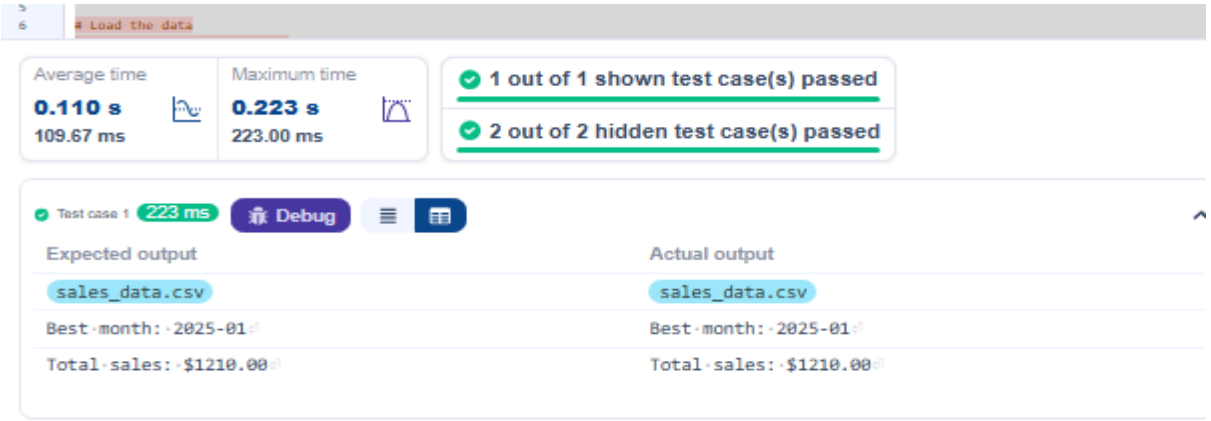
Program :

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Convert the Date column to datetime format
df['Date'] = pd.to_datetime(df['Date'])

# Add a Year-Month column by formatting the Date column
df['YearMonth'] = df['Date'].dt.to_period('M')

# Calculate total sales for each row
df['Total_Sales'] = df['Quantity'] * df['Price']

# Group the data by Year-Month and calculate total sales for each month
monthly_sales = df.groupby('YearMonth')['Total_Sales'].sum()

# Find the month with the highest total sales
best_month = monthly_sales.idxmax()
highest_sales = monthly_sales.max()
```

print(f"Best month: {best_month}")

print(f"Total sales: ${highest_sales:.2f}")

Output :



---

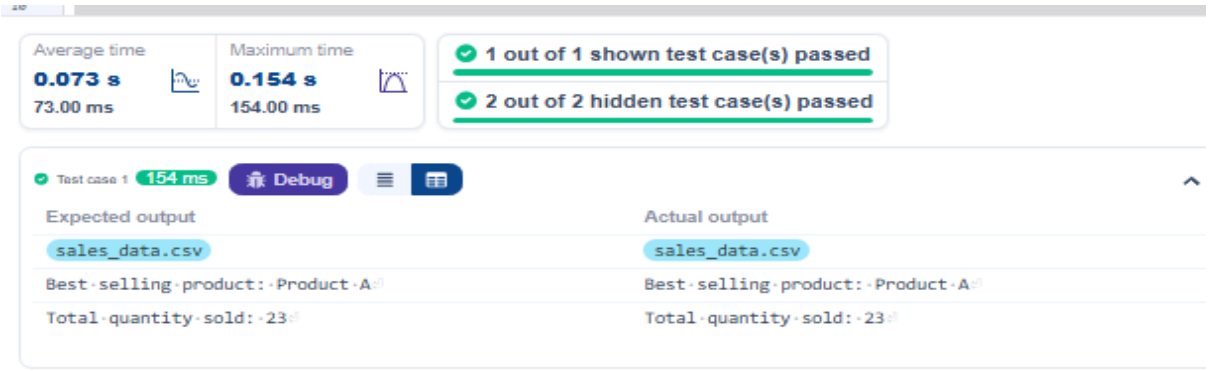
5)

Program :

```
import pandas as pd


# Prompt the user for the file name

file_name = input()


# Load the data

df = pd.read_csv(file_name)


product_sales = df.groupby('Product')['Quantity'].sum()
```

```
# Find the product with the highest total quantity sold

best_product = product_sales.idxmax()

highest_quantity = product_sales.max()



# Display the result

print(f"Best selling product: {best_product}")

print(f"Total quantity sold: {highest_quantity}")
```

Output :



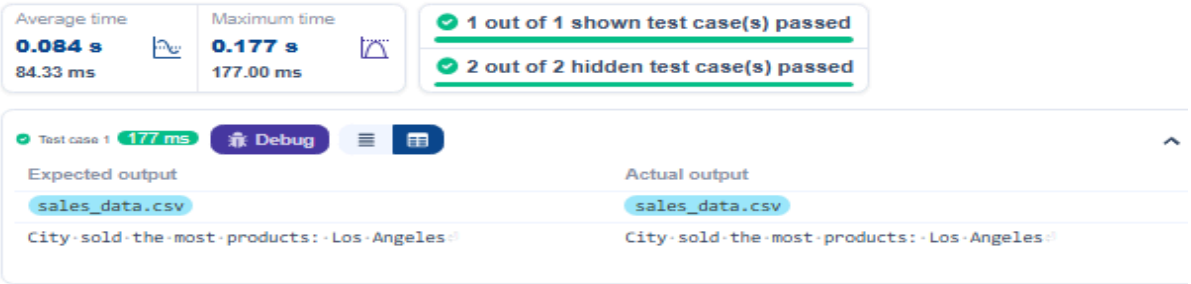
6)

Program :

```
import pandas as pd


# Prompt the user for the file name
```

```
file_name = input()


# Load the data

df = pd.read_csv(file_name)


city_sales = df.groupby('City')['Quantity'].sum()


# Find the city that sold the most products

best_city = city_sales.idxmax()

highest_quantity = city_sales.max()

# write the code..



# Display the result

print(f"City sold the most products: {best_city}")
```

Output :

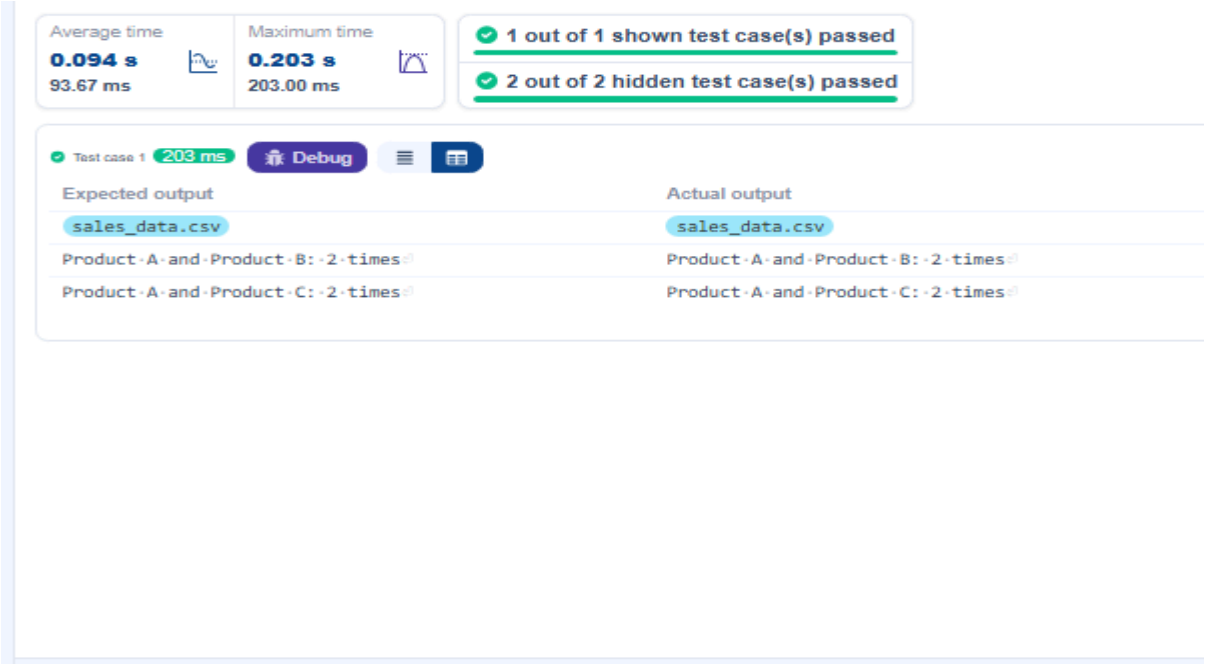


7)

Program :

```
import pandas as pd

from itertools import combinations

from collections import Counter


# Prompt user to input the file name

file_name = input()


# Read data from the specified CSV file

df = pd.read_csv(file_name)


count_pairs=Counter()

# write the code


for date,group in df.groupby("Date"):

        products = group['Product']

        for pair in combinations(sorted(products), 2):

                count_pairs[pair] +=1


# Output the most frequent product pairs


best_count= max(count_pairs.values())

best_pair= [pair for pair,val in count_pairs.items() if val==best_count ]

for pair in best_pair:

        print(f"{pair[0]} and {pair[1]}: {best_count} times")
```

Output :



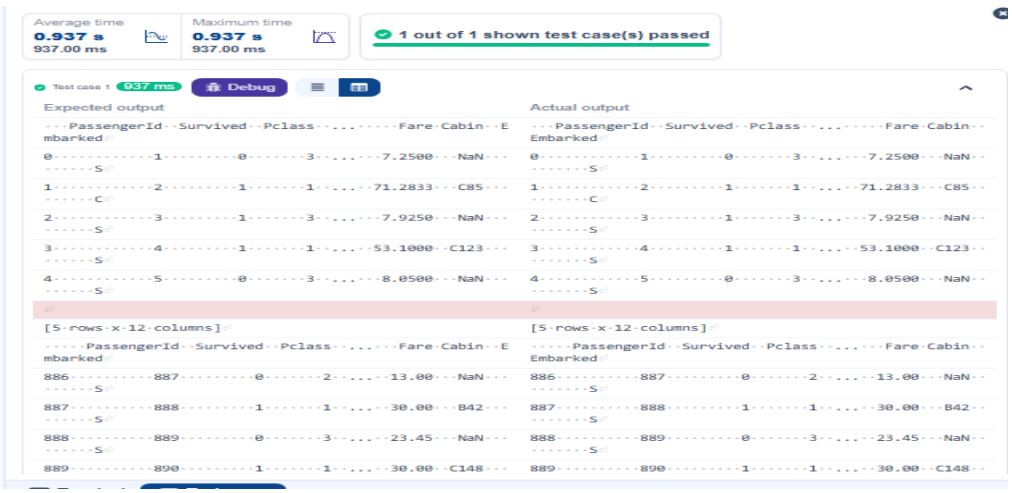
8)

Program :

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset

print(data.head(5))


# 2. Display the last 5 rows of the dataset

print(data.tail(5))


# 3. Get the shape of the dataset

print(data.shape)
```

```
# 4. Get a summary of the dataset (info)

print(data.info())


# 5. Get basic statistics of the dataset

print(data.describe())


# 6. Check for missing values


print(data.isnull().sum())

# 7. Fill missing values in the ‘Age’ column with the median age



# 8. Fill missing values in the ‘Embarked’ column with the mode



# 9. Drop the ‘Cabin’ column due to many missing values



# 10. Create a new column 'FamilySize’ by adding ‘SibSp' and ‘Parch'
```

Output :



9)

Program :

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

df = pd.DataFrame(data)

# 1. Create a new column ‘IsAlone' (1 if alone, 0 otherwise)

df['FamilySize'] = df['SibSp'] + df['Parch']

df['IsAlone'] = (df['FamilySize'] == 0).astype(int)

# 2. Convert ‘Sex' to numeric (male: 0, female: 1)

df['Sex'] = df['Sex'].map({'male' : 0, 'female' : 1})


# 3. One-hot encode the ‘Embarked' column

df = pd.get_dummies(df, columns=['Embarked'])


# 4. Get the mean age of passengers
```

```
mean_age = df['Age'].mean()

# 5. Get the median fare of passengers
median_fare = df['Fare'].median()

# 6. Get the number of passengers by class
passengers_by_class = df['Pclass'].value_counts()

# 7. Get the number of passengers by gender
passengers_by_gender = df['Sex'].value_counts()

# 8. Get the number of passengers by survival status
passengers_by_survival = df['Survived'].value_counts()

# 9. Calculate the survival rate
survival_rate = df['Survived'].mean()

# 10. Calculate the survival rate by gender
survival_rate_by_gender = df.groupby('Sex')['Survived'].mean()
print(mean_age)
print(median_fare)
```

```
print(passengers_by_class)

print(passengers_by_gender)

print(passengers_by_survival)

print(survival_rate)

print(survival_rate_by_gender)
```

Output :

0.878 s
878.00 ms

0.878 s
878.00 ms

1 out of 1 shown test case(s) passed

Test case 1 878 ms Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3 491	3 491
1 216	1 216
2 184	2 184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0 577	0 577
1 314	1 314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0 549	0 549
1 342	1 342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3838383838383838	0.3838383838383838

10)

Program :

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
```

```
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

df = pd.DataFrame(data)

# 1. Calculate the survival rate by class

grouped = df.groupby('Pclass')['Survived'].mean()

print(grouped)

# 2. Calculate the survival rate by embarked location

print(df.groupby('Embarked_S')['Survived'].mean())


# 3. Calculate the survival rate by family size

print(df.groupby('FamilySize')['Survived'].mean())


# 4. Calculate the survival rate by being alone

print(data.groupby('IsAlone')['Survived'].mean())


# 5. Get the average fare by class

print(data.groupby('Pclass')['Fare'].mean())


# 6. Get the average age by class

print(data.groupby('Pclass')['Age'].mean())


# 7. Get the average age by survival status

print(data.groupby('Survived')['Age'].mean())


# 8. Get the average fare by survival status

print(data.groupby('Survived')['Fare'].mean())


# 9. Get the number of survivors by class
```

```
print(data[data['Survived']==1]['Pclass'].value_counts())
```

```
# 10. Get the number of non-survivors by class

print(data[data['Survived']==0]['Pclass'].value_counts())
```

Output :

```
import pandas as pd
```

Average time
0.394 s
394.00 ms

Maximum time
0.394 s
394.00 ms

1 out of 1 shown test case(s) passed

Test case 1 394 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 0.506073	0 0.506073
1 0.336957	1 0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0 0.303538	0 0.303538
1 0.552795	1 0.552795
2 0.578431	2 0.578431
3 0.724138	3 0.724138
4 0.200000	4 0.200000
5 0.136364	5 0.136364
6 0.333333	6 0.333333
7 0.000000	7 0.000000

11)

Program :

```
import pandas as pd

import numpy as np
```

```
# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

df = pd.DataFrame(data)

# 1. Get the number of survivors by gender

print(df[df['Survived']==1]['Sex'].value_counts())

# print((df.groupby('Sex')['Survived']==1).value_counts())

# 2. Get the number of non-survivors by gender

print(data[data['Survived']==0]['Sex'].value_counts())


# 3. Get the number of survivors by embarked location

print(data[data['Survived']==1]['Embarked_S'].value_counts())


# 4. Get the number of non-survivors by embarked location

print(data[data['Survived']==0]['Embarked_S'].value_counts())


# 5. Calculate the percentage of children (Age < 18) who survived

print(data[data['Age']<18]['Survived'].mean())


# 6. Calculate the percentage of adults (Age >= 18) who survived

print(data[data['Age']>17]['Survived'].mean())


# 7. Get the median age of survivors

print(data[data['Survived']==1]['Age'].median())


# 8. Get the median age of non-survivors
```

```
print(data[data['Survived']==0]['Age'].median())
```

```
# 9. Get the median fare of survivors

print(data[data['Survived']==1]['Fare'].median())
```

```
# 10. Get the median fare of non-survivors

print(data[data['Survived']==0]['Fare'].median())
```

Output :

Average time
0.383 s
383.00 ms

Maximum time
0.383 s
383.00 ms

1 out of 1 shown test case(s) passed

Test case 1 383 ms Debug

Expected output	Actual output
female····233	female····233
male·····109	male·····109
Name:·Sex,·dtype:·int64	Name:·Sex,·dtype:·int64
male·····468	male·····468
female····81	female····81
Name:·Sex,·dtype:·int64	Name:·Sex,·dtype:·int64
1····217	1····217
0····125	0····125
Name:·Embarked_S,·dtype:·int64	Name:·Embarked_S,·dtype:·int64
1····427	1····427
0····122	0····122
Name:·Embarked_S,·dtype:·int64	Name:·Embarked_S,·dtype:·int64
0.5398230088495575	0.5398230088495575
0.3810316139767055	0.3810316139767055
28.0	28.0
28.0	28.0
26.0	26.0
10.5	10.5